technicians all require controlled access, but none of them belong in ACME's internal directory. Creating accounts for these groups is costly and challenging to manage at scale.

To address some of these needs, ACME experimented with federation. Several large suppliers and clinic networks were onboarded through SAML or OIDC connections, which worked adequately when both organizations had the technical maturity and ongoing interest to manage a bilateral relationship. However, the limitations soon became clear. Federation requires each partner to operate an identity provider. It also requires ACME to configure that provider, manage metadata, coordinate attribute mappings, and troubleshoot failures that occur outside its administrative boundary. More importantly, federation does not work for many of the organizations ACME interacts with. Small contractors, regional clinics, and field-service companies often lack the infrastructure or motivation to support it.

At the same time, ACME's authorization requirements need more flexibility. Decisions increasingly rely on information from outside ACME's systems: professional certifications, clinical affiliations, project-specific delegations, and supplier relationships, to name a few. These are authoritative statements, but ACME is not the authority that can issue or maintain them. The company needs a way to trust information held by customers, partners, licensing agencies, and other external organizations without importing all that identity data into its own systems.

ACME also faces a growing lifecycle burden. Temporary accounts have multiplied. Short-term access often lasts longer than planned because deprovisioning depends on manual signals from partners about when work is done or personnel changes. These signals often arrive late or not at all. ACME cannot query a partner's HR system to know if a contractor has left or a technician has changed roles, so temporary access often exceeds the circumstances that justified it.

This chapter introduces a different approach. As we discussed in chapter 2, decentralized identity models shift trust to the parties who can legitimately assert these facts, rather than force everything through a central directory or federated identity provider. ACME has built its authorization engine to use attributes provided by verifiable credentials. *Verifiable* credentials are